CHE 301 GDB solution Spring 2021

Which technique do you think is more efficient among Gas Chromatography & Fractional Distillation and why?

Fractional distillation is the separation of components depends upon the boiling point. I think Gas chromatography is more efficient than fractional distillation because of the use of theoretical plate. These theoretical plates don't used in the fractional distillation. Due the increase the number of these plate it can better separate the components of mixture. Gas chromatography uses relatively long packed or open tubular capillary columns and is subsequently far more efficient at separation than fractional distillations with short reflux columns. In addition, gas chromatography uses packing or stationary phases that can be liquid or solid and may exhibit an affinity toward the compounds being separated. The column efficiency of a gas chromatography column is gauged by the number of theoretical plates. The concept of a plate is a carry-over from the first fractionating columns which used discrete plates for separation. The chromatography column does not have discrete plates.

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